

Goal document

Working Title, Personnel & Schedule

- **Working Title:** Evaluating Performance and Generalization of LLM-Synthesized Supervisory Control for a Grinding Mill Circuit
- **Students:** Otto Mörner (M.Sc. Automation & Control, LTH)
- **Industrial Partner:** Weir Group
- **LTH Supervisor:** Richard Pates
- **Timeline:** Autumn Semester 2026 (4 Months)

Background, Context & Motivation

Continuous processes in mineral processing run around the clock and must deliver a product of consistent quality while respecting equipment and safety constraints. They are strongly non-linear and tightly coupled, and several key process states are only indirectly measured. Control solutions must also be transparent enough for operators and engineers to trust and maintain.

A classic control problem in the mineral industry is mill circuit control, where the grinding mill, sump and classification unit (i.e., cyclone clusters) form a coupled process that determines product particle size and circuit operating conditions. The control objective is to regulate product P80 around a specified target, where P80 represents the particle size below which 80% of the product mass falls. In addition, the controller should satisfy operating constraints, such as stabilising the sump level and maintaining the circulating load within acceptable operating limits. This task is challenging because of non-linear process dynamics, interactions between controlled variables, and different response times across components in the circuit. In a real-world mining plant, variations in fresh feed mass flow, ore density and percentage of solids further change the operating conditions and affect grinding and classification performance. Consequently, a control action that improves particle size regulation may adversely affect sump level or circulating load. The supervisory controller must therefore coordinate setpoint adjustments to satisfy these competing requirements, reject process disturbances, and maintain performance under operating conditions beyond those used during controller development.

In practice, grinding circuits use PID loops whose setpoints are adjusted by operators, expert systems or fuzzy-logic supervisors. These build on established operating knowledge but require considerable tuning, and hand-crafted rules grow complex as new conditions are added. Model predictive control and, more recently, learning-based approaches that combine learned world models with control address coordination through prediction and optimisation, but their decisions are hard to inspect and their behaviour beyond the training data is uncertain. This leaves a gap between supervisory logic that is interpretable but costly to maintain, and controllers that are adaptive but opaque.

Large language models (LLMs) have recently emerged as a new paradigm in control and optimisation, as they can generate and iteratively refine code and heuristics from environment feedback (e.g., [PromptPO](#), [Learning Beyond Gradients](#)). Using an LLM directly in the control loop is unsuitable because of latency, non-determinism and safety risks. This project therefore uses the LLM as a meta-supervisor that synthesises explicit, interpretable supervisory rules, which run locally and update the setpoints of the existing low-level controllers. The proposed solution will be benchmarked against Weir's current TD-MPC-type recommendation algorithm, which combines learned world-model prediction with control.

Overall Goals, Research Questions & Hypothesis

The objective is to evaluate how well LLM-synthesised supervisory heuristics coordinate setpoints in a grinding circuit compared with a model-based predictive supervisory controller, and to identify where their performance degrades.

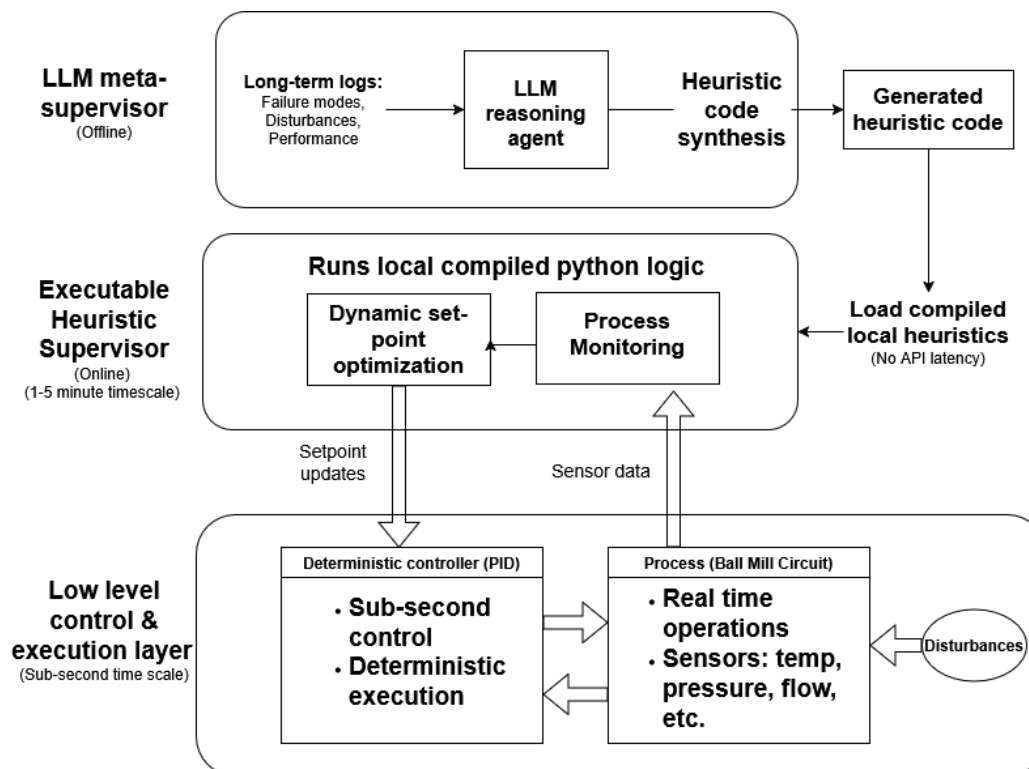
1. **RQ1 (Comparative performance):** How does an LLM-synthesised supervisory heuristic compare with a model-based predictive supervisory controller in mill circuit control, in terms of P80 regulation, sump level stability, compliance with circulating load limits and smoothness of setpoint adjustments, under nominal conditions and disturbances within the development range?
2. **RQ2 (Robustness and generalisability):** How does this performance change when the amplitudes, temporal patterns or combinations of disturbances fall outside the development range, and what failure modes and empirical operating limits can be identified?

The research questions are tested through the following null hypotheses:

- $H_{0,1}$: Within the development range, there is no difference in performance metrics between the LLM-synthesised supervisor and the benchmark controller.
- $H_{0,2}$: Beyond the development range, the degradation in performance relative to the development range is the same for both controllers.

Methodology & Approach

The framework decouples real-time, high-frequency execution from high-level reasoning across three operational layers:



1. **LLM Meta-Supervisor (Heuristic Synthesis Layer):** Uses prompt-based reasoning to generate, update, and maintain explicit, human-readable supervisory code/heuristics (e.g., Python rule sets or decision parameters) based on aggregated plant state summaries and diagnostic feedback. This is the only layer making calls to the API.
2. **Executable Heuristic Supervisor (Real-Time Supervisory Loop):** Executes the compiled, LLM-generated heuristic logic on a 1–5 minute timescale to evaluate aggregated telemetry streams, diagnose operational anomalies, and dynamically update target setpoints. This, and the first layer would replace the current TD-MPC supervisory controller.
3. **Deterministic Low-Level Control Loop:** Executes sub-second PID commands directly on the dynamic grinding circuit simulator using pre-existing industrial machine controllers. This layer will not be changed by this project.

All experiments are carried out in Weir’s dynamic grinding circuit simulator. The benchmark is Weir’s TD-MPC implementation, a model-based reinforcement learning controller that uses a learned latent dynamics model to plan supervisory setpoint adjustments. Both controllers use the same measurement and action interfaces, low-level loops, action limits and test scenarios, and the benchmark’s training coverage and tuning will be documented. Existing expert or fuzzy-logic rules may be included as an additional baseline if available.

Design of experiments

The experiment has two independent sources of randomness: the LLM’s code generation and the simulated disturbances. To separate them, several supervisory policies will be generated independently (separate generation runs from the same prompt setup), and every policy, together with the benchmark, will be evaluated on the same set of seeded scenarios with matched initial conditions and disturbance sequences. Variation between policies then shows how reliable the generation process is, not only how good a single policy is. Each scenario acts as a block, allowing paired comparisons between controllers.

The experiment varies four factors: the controller (LLM-synthesised, TD-MPC and an optional rule-based baseline), the disturbance range (within or beyond the development range), the disturbance type (fresh feed flow, ore density, percentage of solids, and combinations of these) and the temporal pattern of the disturbance (e.g., steps, ramps and oscillations). N independently generated LLM policies will each be evaluated on M seeded scenarios for every factor combination, with N and M chosen after pilot runs based on the observed variance.

The measured responses are metrics for P80 tracking, sump level stability, circulating load compliance and setpoint smoothness, for example integrated absolute error, recovery time, constraint violations and setpoint total variation, together with rule count as a measure of the complexity of the generated logic. Exact definitions, tolerances and evaluation windows will be fixed before testing.

$H_{0,1}$ will be tested with paired comparisons for each metric within the development range (paired t-tests, or Wilcoxon signed-rank tests where normality cannot be assumed). $H_{0,2}$ will be tested with an ANOVA or a mixed-effects model that tests the interaction between controller and disturbance range, treating the generated policy as a random effect. Effect sizes and confidence intervals will be reported at a significance level of 0.05.

To strengthen external validity, scenarios will cover several operating points and combinations of disturbances, the prompts and data summaries given to the LLM will be documented, and the frozen policies will be tested on disturbance scenarios that were never shown to the LLM during generation.

Scientific Foundation & Academic Basis

The thesis draws on three bodies of work:

- Supervisory control of grinding circuits: expert systems, fuzzy logic and MPC in mineral processing.
- Learning-based control: model-based reinforcement learning and world-model methods for process control (e.g., TD-MPC).
- LLMs for control and optimisation: prompt-based, gradient-free optimisation and heuristic learning frameworks (e.g., [PromptPO](#), [Learning Beyond Gradients](#)), applied here to supervisory code synthesis.

Weir will provide the existing implementation and onboarding support, allowing the project to focus on heuristic synthesis and evaluation rather than rebuilding the benchmark.

Expected Contribution to Knowledge

Scientific contributions, tied to the research questions:

- **C1 (RQ1):** A quantified comparison, with effect sizes and confidence intervals, of an LLM-synthesised supervisor and a TD-MPC controller on P80 regulation, sump level, circulating load and setpoint smoothness, tested through $H_{0,1}$.
- **C2 (RQ2):** Degradation curves showing how each controller's performance changes as disturbances move beyond the development range, tested through $H_{0,2}$, together with a catalogue of observed failure modes and an empirical operating envelope for the LLM-based supervisor.
- **C3:** An estimate of how much performance varies between independently generated policies, indicating how reliable LLM-based synthesis is as a design method.

Practically, the results will show Weir when inspectable supervisory heuristics are a viable alternative and where predictive control remains advantageous. Negative or inconclusive results will also be reported.

Delimitations: the study is limited to one grinding circuit in simulation. Formal stability guarantees, plant deployment and sensor-fault diagnosis are outside the scope.

Resources & Feasibility

Workspaces will be provided by Weir Group alongside LTH campus facilities. The dynamic grinding circuit simulator will run on local computing resources. Necessary LLM API access and compute infrastructure for heuristic synthesis and real-time supervisory evaluation will be secured in collaboration with Weir Group.